

INTRODUCTION TO DEMOGRAPHY

WHAT DEMOGRAPHY IS

- **Scientific study of the size, composition (or structure) of and distribution of the human population through the interaction of fertility, mortality and migration.**
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- **Because it involves the analysis and manipulation of numbers, demography makes considerable use of mathematical and statistical tools.**
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- **Demographic is not static but dynamic for it also studies population change in terms of determinants and consequences.**
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- **It is also concerned with the spatial distribution of the population.**
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WHAT DEMOGRAPHY IS

- **Demography therefore concerns itself with :**
- **Population size**
- **Population structure or composition**
- **Population distribution**
- **Population change due to births, deaths, movements.**

POPULATION SIZE

- **This is the number in absolute terms of people within a defined territory or space.**

POPULATION SIZE

	Population size	Annual growth rate	Population size	Annual growth rate	Population size	Annual growth rate
	1980	1969-1980	1990	1980-1990	2000	1990-2000
Zambia	5,661,801	3.1	7,383,097	2.7	9,885,591	2.4

POPULATION SIZE

- The population of a province, district, city, or township or country is often given in terms of an absolute count. Population size can be expressed in relative terms as percentage.

WHAT DEMOGRAPHY IS

Central	1,012,257	10.2
Copperbelt	1,581,221	16.0
Eastern	1,306,173	13.2
Luapula	775,353	7.8
Lusaka	1,391,329	14.1
Northern	1,258,696	12.7
Northwestern	583,350	5.9
Southern	1,212,124	12.3
Western	765,088	7.7
Total	9,885,591	100.0

POPULATION COMPOSITION

- **This refers to the constituent elements (parts, fractions or segments) that make up a population. These are expressed in absolute or relative terms.**
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- **The major elements of a population are:**
 - **Age**
 - **Sex**
 - **Marital status**
 - **Education**
 - **Employment status**
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POPULATION COMPOSITION

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- **According to the 2000 census, 51 percent of the population was female while 49 percent was male.**
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- **In 2000, 45.3 percent of the population was under 15 years (ie, between 0-14 years)**

POPULATION GROWTH (OR CHANGE)

- **POPULATION GROWTH (OR CHANGE)**
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- **This refers to increase or decrease in absolute or relative terms of the population.**
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- **This is expressed as percent change or the annual population growth rate of growth.**
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- **Population growth can be geometric**
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- **It can also be exponential.**

POPULATION GROWTH (OR CHANGE)

- **EXAMPLE**

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- **The Zambian population**
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- **1995 – 9,112m**

- **2000- 10,722m**
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- **Percent change = $10,722 - 9,112 / 9,112$**

- **18 percent**
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- **The population increased by 1,610 or 17.7 percent.**

POPULATION GROWTH (OR CHANGE)

	Population size	Annual growth rate	Population size	Annual growth rate	Population size	Annual growth rate
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POPULATION DISTRIBUTION

- This refers to the location of the population in space or settlement patterns and the dispersion or concentration of the population in specific areas.
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- This often expressed in terms of residence such as rural and urban residence.
- Population can also be distributed spatially within provinces and districts.

POPULATION DISTRIBUTION

Thirty eight percent of the population in Zambia is urban. Sixty percent is rural.

Residence in a given space determines population density.

POPULATION DISTRIBUTION

Rural	5990356	60.6
Urban	3347069	33.9
Total	9337425	94.5

POPULATION DISTRIBUTION

- RURAL URBAN POPULATION IN ZAMBIA (2000)

Rural	5,990,356	60.6
Urban	3,347,069	33.9
Total	9,337,425	94.5

POPULATION DISTRIBUTION

- This show that thirty eight percent of the population in Zambia is urban whilst sixty percent is rural.

POPULATION DISTRIBUTION

- Population distribution can also be expressed in terms of population density.
- Residence in a given space determines population density
- Population density is expressed as the number of people per unit of land area.
- Density is reported in terms of number of people per square kilometre.
- $PD = \text{Population} / \text{Area}$

POPULATION BY PROVINCE

PROVINCE	POPULATION	
CENTRAL	1,267,803	
COPPERBELT	1,958,623	
EASTERN	1,606,319	
LUAPULA	958,976	
LUSAKA	2,198,996	
MUCHINGA	743,109	
NORTHERN	1,117,903	
NORTHWESTERN	706,462	
SOUTHERN	1,606,793	
WESTERN	881,524	
TOTAL	13,046,508	

POPULATION BY DISTRICT

DISTRICT	POPULATION
LUSAKA	1,742,979
KITWE	522,092
NDOLA	455,194
MAZABUKA	261,268
KAPIRI-MPOSHI	240,841
CHINGOLA	210,073
KABWE	202,914
MUFULIRA	161,601
LUANSHYA	153,117
LIVINGSTONE	142,034

POPULATION DISTRIBUTION

- POPULATION DENSITY BY PROVINCE 2010

PROVINCE	DENSITY
Lusaka	100.4
Copperbelt	62.5
Eastern	24.6
Luapula	19.0
Southern	18.8
Central	13.4
Northern	11.9
Western	7.0
North-western	5.6

POPULATION DISTRIBUTION

- TEN MOST DENSELY POPULATED TOWNS AND CITIES IN ZAMBIA 2010

Lusaka	4,841.6
Kitwe	671.9
Ndola	412.7
Livingstone	204.4
Luanshya	188.8
Kalulushi	132.7
Kabwe	129.1
Chingola	125.2
Mufulira	98.7
Chililabombwe	88.2

POPULATION DISTRIBUTION

- TEN LEAST DENSELY POPULATED TOWNS AND CITIES IN ZAMBIA 2010

Mufumbwe	2.7
Kasempa	3.2
Sesheke	3.2
Itezhi-tezhi	4.0
Mpika	5.2
Lukulu	5.2
Chama	5.8
Kazungula	5.8
Shang'ombo	5.9
Zambezi	6.3

POPULATION CHANGE

- **Change in the size, composition, and distribution of the population results from the interaction of three important demographic variables – fertility, mortality, and migration.**

POPULATION CHANGE

- **FERTILITY**

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- This refers to the number of births occurring in a population.

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- **MORTALITY**

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- This refers to the number of deaths occurring in a population.

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- **MIGRATION**

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- This refers to the number of people moving in and out of the population.

POPULATION CHANGE

- This process of population change is often depicted through the balancing equation.
- This presents population change as a function of the initial population and the interaction of fertility (births), mortality (deaths), and migration (movements in and out of the population) in the intervening period.
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- $P_t = P_o + B - D + I - O$
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- where P_t = Current population
- P_o = Previous population
- D = Deaths between current and previous population
- B = Births between current and previous population
- I = In-migrants between current and previous population
- O = Out-migrants between current and previous population

POPULATION CHANGE

- For example, the population of 2020 can be expressed as a function of the population of 2010 and the demographic events between 2010 and 2020.
- $P_{2020} = P_{2010} + B_{2010-2020} - D_{2010-2020} + I_{2010-2020} - O_{2010-2020}$

FORMAL DEMOGRAPHY

- **This is the more technical side of demography, involving the use mathematics and involving a lot of computations.**
- **This involves dealing with demographic variables mathematically.**
- **The collection and statistical analysis of demographic data is one example.**

FORMAL DEMOGRAPHY

- **EXAMPLES OF FORMAL DEMOGRAPHY**
- **Execution and administration of censuses**
- **Computation of life expectancy using life tables**
- **Projection of the population**
- **Estimation of mortality and fertility using incomplete data.**
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- **More concerned with relationships among demographic variables themselves.**
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- **Using model life tables to see how changes in fertility and mortality influence changes in the age structure of the population.**

POPULATION STUDIES

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- **The interdisciplinary approach to demography involving biological and social sciences.**
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- **EXAMPLES OF POPULATION STUDIES**
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- **Studies of the relationship between demographic and non- demographic variables**
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POPULATION STUDIES

- **1. Impact of non- demographic on demographic variables**
- **Changes in income – fertility or mortality**
- **Changes in education – fertility or mortality**
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- **2. Impact of demographic variables on non-demographic variables**
- **Fertility – change in consumption patterns**
- **Aging of population – change in voting patterns.**
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REASONS FOR STUDYING POPULATION

- **Demography enables us to do the following:-**
 - 1. Determine the number and distribution of people in a given area and to know the profile or composition by age, sex, marital status, employment status, occupation, educational level, etc.**
 - 2. Analyze the role of economic and social factors in influencing population change.**
 - 3. Analyze the role of demographic factors -fertility, mortality, migration in influencing social and economic change.**

REASONS FOR STUDYING POPULATION

4. Determine the relationship between population change and socio-economic change.

- What happens to the economy or society when there is change in the population? Or what happens to the population when there is a change in the economy or society?

• 5. Predict population changes and their consequences.

- Based on past trends in fertility, mortality, migration one can forecast future population trends. On this basis, it is also possible to envisage what is going to happen – will this result in overpopulation in relation to available resources for example?



REASONS FOR STUDYING POPULATION

- **6. Demographic knowledge assists both private and public organizations to plan and make policies aimed at improving the welfare of the entire population.**

REASONS FOR STUDYING POPULATION

- **7. Demographic knowledge is indispensable in the following areas:**
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 - **Planning for the provision of education**
 - **Planning for provision of health facilities,**
 - **Planning for provision housing units,**
 - **Setting of targets in agricultural and industrial production.**
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HISTORY AND DEVELOPMENT OF DEMOGRAPHY

- **John Graunt** regarded as **FATHER of DEMOGRAPHY**.
- Others argue that it was **Ibn Khaldun (1332-1406)** a Muslim Scholar is the "father of demography" as he produced the first scientific and theoretical work on population, development, and group dynamics.
- **Graunt, however, did more of what can be considered serious demographic work.**

JOHN GRAUNT (1620 – 1674)

- He pioneered the **NUMERICAL ANALYSIS** of **BIRTHS** and **DEATHS** using **DATA** from **BILLS** of **MORTALITY** – information on burials and christenings compiled by parish clerks initiated during the great plagues of the 16th and 17th centuries.
- He used this data to attempt a **DISCOVERY** of **REGULARITIES** and **UNIFORMITIES** (i.e. **COMMON PATTERNS**) in **VITAL EVENTS** (births, deaths, marriages, etc).
- On this basis he attempted to formulate the laws of population growth in a book entitled **NATURAL AND POLITICAL OBSERVATIONS ...made upon the Bills of Mortality** (1662).

JOHN GRAUNT (1620 – 1674)

- **HIS DISCOVERIES**

- **In London, deaths, exceeded births – lack of growth.**
- **In rural areas, births exceeded deaths – high growth – migration.**
- **At birth, number of male infants exceeded female infants.**
- **Established a distributive age pattern to deaths.**

JOHN GRAUNT (1620 – 1674)

- Classified deaths by a cause and geographical variation of causes and by year.
- Constructed a primitive and crude form of a life table.
- Estimated of London's population.
- Identified correctly migration, fertility, and mortality as components of population growth.
- Pioneered the use of statistical methods in evaluating demographic data

EDMOND HALLEY

- He was a mathematician who improved on the work of Graunt and developed the life table as the basis for life insurance mathematics in 1693.
- Richard Price was credited with the first textbook on life contingencies published in 1771, using life tables extensively.
- He was followed later by Augustus de Morgan, 'On the Application of Probabilities to Life Contingencies', (1838).¹

THOMAS MALTHUS (1766 – 1834)

- British clergyman and economist and one of the pioneers of demography.
- He wrote a book an ESSAY ON POPULATION in 1798.
- The book was influences by the following events in Europe:
- The outbreak of the French Revolution which scared the upper classes in other European countries.
- The British discussion on the improvements of social security for the poor.

Thus the book stressed the need to curb the population of the poor.

THOMAS MALTHUS (1766 – 1834)

- **MALTHUS' BASIC ARGUMENT**

- His basic argument was that food production increased by arithmetic progression (very slowly) – 1, 2, 3, 4, 5, 6
- Population on the other hand increased by geometric progression (very fast) 1, 2, 4, 8, 16, 32.
- Thus population was doubling every 25 years.
- The rapid increase in population was according to him was attributable to the reproductive habits of the poor who “bred like animals.”
- Since this endangered the food supply, he suggested the need to curb the population growth of the poor.

THOMAS MALTHUS (1766 – 1834)

- Malthusian solution to the problem of rapid population growth.
- His moral solution was that the poor should restrain their sexual life by late marriages and abstinence.
- He further identified two checks on population growth.
 - Preventive checks
 - Positive checks

THOMAS MALTHUS (1766 – 1834)

- **PREVENTIVE CHECKS**
- These are more voluntary mechanisms (or means) of controlling births. These include the following:-
- **PROLONGED CELIBACY** – Abstention from marriage.
- **CHASTITY** – Abstention from all sexual intercourse.
- Excluded from these were:
 - **Adultery**
 - **Prostitution**
 - **Sexual deviations** (lesbianism/homosexuality – christened vicious.
 - **Birth control**
 - **Abortion**

THOMAS MALTHUS (1766 – 1834)

- **POSITIVE CHECKS**

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- **These included involuntary checks such as:**

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- **Wars**
- **Epidemics – AIDS etc**
- **Famines**
- **Food shortage – the ultimate positive check.**
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- **His argument was that if fertility could not be controlled through preventive checks, positive checks were inevitable leading to high mortality thus curbing population growth.**

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THOMAS MALTHUS (1766 – 1834)

- He discouraged improvement of the social conditions of the poor as this would encourage them to have more children.
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- Thus he advocated for the reduction of the population of the poor through preventive checks.
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- On the other hand, he argued that the demand for food, was an important incentive for the economy.
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- He therefore, advocated that the upper classes should expand their consumption in order to create this incentive.
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- Malthus is seen as the intellectual father of ideas of overpopulation and the limits to growth.
- Later more sophisticated and realistic models were presented by e.g. Benjamin Gompertz.

KARL MARX (1818-1883) AND FRIEDRICH ENGELS (1820-1895)

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- **Marx and Engels did not view population pressure as constraint to economic growth.**
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- **They thus rejected the Malthusian perspective.**
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- **They believed that each economic system has its own laws of population.**
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- **In the capitalist mode of production, population could be changed to a surplus population or an industrial reserve army.**

KARL MARX (1818-1883) AND FRIEDRICH ENGELS (1820-1895)

- **Three forms of surplus population**
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- **Floating population – displaced by machinery**
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- **Latent population – agricultural population (potential migrants) displaced by commercialization and mechanization**
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- **Stagnant population – those with highly irregular employment – lowest living standards.**

KARL MARX (1818-1883) AND FRIEDRICH ENGELS (1820-1895)

- **In this way, poverty was as a result of the evils of such a social organization as it was possible to manipulate conditions of service by the capitalists by creating artificial unemployment of labour and pushing down wages.**
- **In the socialist mode of production, however, the surplus population could be fully utilized through labour intensive methods of production.**
- **Moreover, in the socialist system, increasing numbers in the population are not a burden to the economy but are potential producers of a future labour force and thus assets.**

WILLIAM PETTY (1623 -1687)

- **Sir William Petty (1623 -1687) was a close and life-long friend of John Graunt's. While Graunt is recognized as the founding father of population studies, Petty stands recognized as the founder of political science. Petty coined the expression "Political Arithmetick" for population and economic studies.**
- **It was not until 1855 that the Frenchman Guillard introduced the designation Demographie.**
- **Petty recommended that every nation should have a statistical office providing government with data for intelligent governance. Although the beginning of demography is 17th century England, Sweden was the first country to implement systematic collection of population data.**

WILLIAM PETTY (1623 -1687)

- **This was accomplished by a royal decree in 1748 which led to the creation of a statistical office in Stockholm in 1749.**
- **Petty tackled, with which demographers and statisticians are still wrestling today, particularly in studies of the problems of under developed countries.**
- **Among other things, he was concerned with population projections, the economics of urbanization, population structure and the labor force, unemployment and under employment, and the measure of national income.”**

OTHER DEVELOPMENT OF DEMOGRAPHY IN THE 19TH CENTURY

During the 19th century, there was growing interest in demography and concern with population problems.

In 1882, demography received explicit international recognition at the 4th International Conference in Hygiene held in Geneva, Switzerland.

In 1885, the International Statistical Institute accorded demography the status of a discipline ,

OTHER DEVELOPMENT OF DEMOGRAPHY IN THE 19TH CENTURY

- The period 1860-1910 was a period of transition when demography emerged from statistics as a separate field of interest.
- The key figures during this transition included the so-called international 'great demographers' like Adolphe Quetelet (1796-1874), William Farr (1807-1883) British epidemiologist regarded as one of the founders of medical statistics.
- Louis-Adolphe Bertillon (1821-1883) – medical doctor and statistician ; and his son Jacques Bertillon(1851-1922) statistician and demographer.

- **Others included:**
- **Joseph Korosi (1844-1906) (Hungarian statistician);**
- **Richard Bockh (1824-1907) (German statistician);**
- **Wilhelm Lexis (1837-1914) German statistician, economist, and social scientist;**
- **Luigi Bodio (1840-1920) an Italian economist and statistician.**
- **These demographers contributed to the development of demography and methods and techniques of demographic analysis.**

DEVELOPMENTS IN THE 20TH CENTURY

- During the 20th century, much of ground – breaking work in demography was done by the following:
- Raymond Pearl (3 June 1879 – 17 November 1940) an American biologist, regarded as one of the founders of biogerontology;
- Lowell Reed (January 8, 1886 – April 29, 1966) was 7th president of the Johns Hopkins University research scientist in biostatistics and public health
- These two refined the application of the logistic curve to human population growth.
- According to this view, population approaches on upper limit as it approaches the full utilization of its resources.

OTHER DEVELOPMENT OF DEMOGRAPHY IN THE 19TH CENTURY

- Further improvements on the logistic curve were made by the following:
- Alfred Lotka (March 2, 1880 – December 5, 1949) an American mathematician, physical chemist, and statistician, famous for his work in population dynamics and energetics;
- George Udny Yule FRS (18 February 1871 – 26 June 1951), a British statistician.
- Other demographers and statisticians made further improvements to the logistic curve.

DEVELOPMENTS IN THE 20TH CENTURY

INSTITUTIONALIZATION OF DEMOGRAPHY

Institutionalization of demography as a separate discipline was pioneered in the USA in the early part of the 20th century.

During the first quarter of the 20th century, a majority of the most competent students of population studies emerged.

These included the following:

Louis Dublin (November 1, 1882 – March 7, 1969) a Jewish American statistician and vice president and statistician of the Metropolitan Life Insurance Company;

Others who played significant roles in these developments included Alfred Lotka, Lowell Reed and Raymond Pearl.

All these demographers were concerned primarily with health, actuarial studies, biological sciences or were these fields.

DEVELOPMENTS IN THE 20TH CENTURY

The scope and orientation of population studies was significantly affected by the entrance of scholars with different backgrounds.

These scholars included others like:

P. K. Whelpton (agricultural economist) and;

Frank Notestein (August 16, 1902 - February 19, 1983) was an American (Economist and Statistician).

This diversity in backgrounds laid the foundation for the interdisciplinary orientation of population studies.

Since the early 1920s, population studies or demography as a field of interrelated interests has grown rapidly.

By 1927, the First World Population Conference was held in Geneva.

Since then, the cooperation between demographers and international organizations had grown.

DEVELOPMENTS IN THE 20TH CENTURY

The first major development in this relationship was the sponsorship by the League of Nations of a series of population studies under the direction of Frank W. Notestein at the Office of Population Research at Princeton University (established in 1936).

In subsequent years, several research and training programs in population studies were established at many universities in the US.

These developments spread to European universities like LSE.

In the early 1950s, the Population Council was established by John D. Rockefeller III.

The Population Council is a foundation devoted to the study and research in the field of population.

DEVELOPMENTS IN THE 20TH CENTURY

The UN through the UNFPA has also contributed significantly to the establishment of research and training institutes.

It has set up several regional training centers in Africa (RIPS), Asia (Bombay), and Latin American (Santiago).

UNFPA has also funded specific training programs at universities such as UNZA.

The program at UNZA was incepted in 1986 and has since produced over 500 graduates with hundred others minoring.

DEMOGRAPHY AND ITS INTERRELATIONSHIP WITH OTHER DISCIPLINES

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- **Demography is a highly interdisciplinary social science.**
- **In other words, it has strong relationship with other disciplines such as economics, sociology, mathematics, geography and many others.**

DEMOGRAPHY AND ITS INTERRELATIONSHIP WITH OTHER DISCIPLINES

- **ECONOMICS**

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- This is a social science concerned with description and analysis of the production distribution, and consumption of goods and services.
- Economics is related to demography in the following ways.
- Economic conditions can influence trends in mortality and fertility.
- For example improvements in income may reduce mortality and fertility levels.
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- In turn, demographic conditions can influence economic trends.
- For example, increase in fertility and reduction mortality may result in younger population which may influence demand for goods and services.

- **Economic models of cost – benefit analysis and the law of supply and demand can be used to explain fertility trends.**
- **For example, at household level, a family decides to have an additional children will look at the cost and benefits of having another child.**
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- **In development planning, population is a critical variable in planning for education, housing development, employment creation, and in constructing quality of life indicator like per capita income.**

DEMOGRAPHY AND ITS INTERRELATIONSHIP WITH OTHER DISCIPLINES

- **SOCIOLOGY**

- **Sociology is concerned study of society, social institutions, and social relationships.**
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- **Sociology is related to demography in the following ways:**
- **Social and cultural conditions can influence trends in fertility, mortality, and migration.**
- **For example, cultural values such as the prestige of having a large family may influence increases in fertility.**
- **Cultural practices such as sexual cleansing may increase the likelihood of HIV and thus mortality.**
- **Cultural practices like circumcision may reduce HIV incidence and reduce mortality.**

DEMOGRAPHY AND ITS INTERRELATIONSHIP WITH OTHER DISCIPLINES

- **SOCIOLOGY**

- In turn, demographic factors can influence social and cultural conditions.
- For example, migration may influence social and cultural change.
- For example, movement of population from Eastern and Northern parts of Zambia resulted in language change in Lusaka and the Copperbelt.
- Colonization also changed cultural values in Zambia.
- Population change such as change in the age profile can influence social change.
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DEMOGRAPHY AND ITS INTERRELATIONSHIP WITH OTHER DISCIPLINES

- **MEDICINE**

- The field of reproductive physiology has improved our understanding of how biological factors may influence human reproduction.
- It describes how breastfeeding may impact on birth intervals thereby influence the number of births a woman can have.
- The medical field is responsible for the development of contraceptives technology which has led to a better way of regulating fertility.

DEMOGRAPHY AND ITS INTERRELATIONSHIP WITH OTHER DISCIPLINES

- **Other related fields in the medical field include:**
- **Morbidity – the study of disease incidence in the population.**
- **Epidemiology - the branch of medicine which deals with the incidence, distribution, and possible control of diseases and other factors relating to health.**
- **For example, analysis of the prevalence of malaria and how this influences death rates.**
- **Biostatistics – the use of the statistical tools in biological, medical, and public health research.**
- **For example analysis of survival rates/death rates of people suffering from cancer declining over time.**

DEMOGRAPHY AND ITS INTERRELATIONSHIP WITH OTHER DISCIPLINES

- **MATHEMATICS**

- **Mathematics and statistics are important tools of analysis of population.**
- **For example, the computation of mortality, fertility, and migration rates relies on the use of basic statistics.**
- **More advanced models like life tables, computer simulations of demographic and economic relationships , and population projections also rely on the use of mathematical and statistical tools.**

- **PSYCHOLOGY**

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- **Psychology is concerned with the mind and human behaviour.**
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- **Human behaviour can influence attitudes towards demographic phenomena like adoption of contraceptives.**
- **For example, emotional inclinations or feelings may influence attitudes towards family planning.**
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DEMOGRAPHY AND ITS INTERRELATIONSHIP WITH OTHER DISCIPLINES

- **GEOGRAPHY**

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- **The focus on the study of the geographical distribution of the population important to geography and city and regional planners is also interest to demographers.**
- **The use of GIS (Geographical Information Systems) to to visually display the relationship between population distribution and resource allocation spatially on maps is of common interest to both demographers and geographers.**

DEMOGRAPHY AND ITS INTERRELATIONSHIP WITH OTHER DISCIPLINES

- **POPULATION ECOLOGY.**
- **This field of study is concerned with populations and how they interact with their environment.**
- **For example, the impact of population increase may lead to environmental degradation.**
- **This may disturb the ecological balance through human settlements and industrialization leading to deforestation and pollution.**
- **Climate change is a result of human pollution and deforestation.**

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DEMOGRAPHY AND ITS INTERRELATIONSHIP WITH OTHER DISCIPLINES

- **GENDER STUDIES**

- The effect of women empowerment and human rights for women in fertility regulation.

- **POLITICAL SCIENCE**

- Social science concerned with the description and analysis of political and especially government institutions and processes.
- It is concerned with distribution of power in society.
- Distribution of power through the electoral process through constituency creation is dependent on population distribution and size.
- Determination of voting age population relies on demographic information.
- Effective campaign strategy is influenced by demographics.

CONTEMPORARY POPULATION PROBLEMS AND ISSUES

- **Lester Brown and his associates at the Worldwatch Institute**
- **Population growth can compound and magnify, if not, create a wide variety of social, economic, and political problem.**
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- **Because of its interdisciplinary nature, demography in population studies deals with diverse issues from other fields.**
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CONTEMPORARY POPULATION PROBLEMS AND ISSUES

FOOD SECURITY

- Food supply must keep pace with population growth.
- If population growth exceeds food supply, a country falls into the Malthusian trap of having more mouths to feed with limited food supplies.
- As a counterweight, a country must have adequate resources required to expand food output – land, water, energy, fertilizer; otherwise famine and starvation may be the outcome.

CONTEMPORARY POPULATION PROBLEMS AND ISSUES

POLLUTION

- Increase in population size increases the potential for disrupting the ecosystem or the – the balance of nature.
- An *ecosystem* includes all of the living things (plants, animals and organisms) in a given area, interacting with each other, and also with their non-living environments (weather, earth, sun, soil, climate, atmosphere).
- In an *ecosystem*, each organism has its' own role to play.
- As population increases, human activities such as industrialization, habitation, mining and deforestation have the potential of disrupting this balance of nature.
- These activities ultimately pose a threat to human life.
- In Zambia, water pollution especially on the Kafue River has increased due to effluents from Nitrogen Chemicals and mines.

CONTEMPORARY POPULATION PROBLEMS AND ISSUES

INFLATION

- **Inflation results when demand exceeds supply or when too much money is chasing too few goods.**
- **Scarcity induced inflation when population increases without corresponding increases in production.**
- **Population increases in the context of a scarcity of goods and services.**

CONTEMPORARY POPULATION PROBLEMS AND ISSUES

INCOME

- **Rapid population growth offsets economic growth.**
- **If the rate of population growth exceeds the rate of economic growth, it will reduce the amount of goods and services available to the population because there will be more people in relation to the available wealth.**
- **All things being equal, this will result in lower per capita income – a standard quality of life measure.**

CONTEMPORARY POPULATION PROBLEMS AND ISSUES

- **UNEMPLOYMENT**

- Increase in population size accelerates growth of the size of the labour force.
- If the increase in the size of the labour force exceeds the economy's capacity to create jobs, the level of unemployment will rise.
- Of course, if the opposite is true such a problem will not arise; there might even be a scenario where there will be more jobs than people available to do them as is the case in some developed countries.

-

CONTEMPORARY POPULATION PROBLEMS AND ISSUES

- **SOCIAL PROBLEMS**

- **Increase unemployment due to rapid population growth may force people to resort to alternative means of survival such as prostitution, crime, etc**

CONTEMPORARY POPULATION PROBLEMS AND ISSUES

LITERACY AND EDUCATION

If population grows faster than the ability of the government to construct schools and provide adequate educational facilities, fewer people are likely to enter the school system.

This will push up the levels of illiteracy and a decline in educational standards.

In developing countries, this has been aggravated by the tidal waves of youths competing for the limited places in the school system.

CONTEMPORARY POPULATION PROBLEMS AND ISSUES

INDIVIDUAL FREEDOM

- **Increase in population size increases demand for land/space.**
- **Land invasions and encroachments are clear manifestation of this phenomenon as squatters invade on other people's land their quest for additional for space.**
- **Land invasions and encroachments represent a violation of the individual freedoms of those whose land has been invaded.**
- **To mitigate this, more rules and regulations are required to supervise individual use of the earth's resources and space for the common good.**

CONTEMPORARY POPULATION PROBLEMS AND ISSUES

•HOUSING AND PROPERTY

- Increase in population induces demand for housing.**
- Consequently, costs of land (property rates), timber, cement and fuel go up beyond the financial capacities of the majority of the population.**
- This has resulted in the construction of substandard houses and the rapid increase in the growth of uncontrolled urban settlements.**
- In Lusaka, it is estimated that more than half of the population lives in these settlements.**

CONTEMPORARY POPULATION PROBLEMS AND ISSUES

WATER AND SANITATION ISSUES

- **Population increase , if not matched by expansion of water and sanitation projects, results in persistent water shortages.**
- **The MCA is attempting to mitigate this in Lusaka.**

CONTEMPORARY POPULATION PROBLEMS AND ISSUES

TRANSPORT

- **Population increase may lead to an increase in labour force and subsequently to the expansion of the moneyed class – the middle class - with more purchasing power to buy more cars.**
- **Increase in vehicular traffic may result in congestion on the congestion on roads.**
- **This may require large scale mass transportation systems, road expansion, flyover roads/overpass, subway systems, etc. to cope with increasing demand for transport.**
- **Other responses may include decentralization of CBDs.**

FLYOVER ROAD



CONTEMPORARY POPULATION PROBLEMS AND ISSUES

•ENERGY

- Increase in population results in demand for energy to prepare food, clothing, and shelter.
- Every person added to the world's population requires energy to prepare food, to provide clothing and shelter and to fuel economic life.
- Each increment in demand leads to shrinkage in energy reserves forcing global adjustments in the use of energy.
- Julian Simon's argument on man's ingenuity as the ultimate resource.

APPLIED DEMOGRAPHY

- **This involves the applications of demographic information and methods in business and public administration.**
- **The involvement is in the following fields.**
 - **Business planning**
 - **- Human resource management**
 - **- Marketing**
 - **- Investment**
 - **- Management demographics**
 - **- Insurance**

APPLIED DEMOGRAPHY

- **Social planning**
-
- **- Education**
- **- Public services**
- **- Health services**
- **Political planning**
- **- Legislative analysis**
- **- Campaign strategizing**

HUMAN RESOURCE MANAGEMENT

- **BUSINESS PLANNING**
- **INTERNAL DEMOGRAPHICS**
 - This is concerned with the understanding, at the micro – level, of the demographic characteristics of institutions or companies.
 -
- **HUMAN RESOURCE MANAGEMENT**
 - Management in organizations have to have knowledge of population composition of its work force in order to maximize productivity and profitability.
 - This is also done in order to adequately respond to the employee's needs because these depend on the employees demographic and non-demographic characteristics.
 - Responding to the employees these needs increases worker morale, motivation, and productivity, and therefore profitability.
 - This requires breaking up the population composition in terms of the demographic and non-demographic variables such as age, sex, marital status, education, skills level, and training.

HUMAN RESOURCE MANAGEMENT

- The demographic profile of a company will influence decision making in the following way:
 -
 - **Age** – Staff development, training, education, and retirement decisions depends on the distribution of the work force by age.
 -
 - **Sex** - Affirmative action policies and staff development of the disadvantaged sex category.
 -
 - **Marital status** - Child care and maternity policies will dependent will dependent on the distribution of work force by marital status and sex.
 -
 - Taking account of the demographic characteristics of the work force in this way may influence levels of productivity and profitability of a company.

HUMAN RESOURCE MANAGEMENT

- The HRM, must understand **EXTERNAL DEMOGRAPHICS**
-
- This refers to the influence of demographic influences outside the organization that may impact on the work force of a company and its operations internally

HUMAN RESOURCE MANAGEMENT

-
- **Fertility, mortality, and migration in the general population outside the company will influence labour force size.**
-
- **This will influence the decline or increase in the size of the labour.**
-
- **In turn, these demographic processes outside the company may influence the internal policies of a company.**
-
- **For example, low fertility and low mortality will result in a decrease the number of young people joining the labour force from institutions of learning.**
-
- **This might result in a shortage of engineers, computers scientists and thus stiff competition for these by companies, resulting in better conditions.**
-

HUMAN RESOURCE MANAGEMENT

- The opposite scenario might result if fertility is high and mortality is low.
- This may also influence retirement policies; older people will be retired since there is a large reservoir of younger people with skills to replace them.
- Making personnel decisions like retirements or recruitment of new staff should take cognizance of the supply of labour outside the organization.
- If there is a shortfall of certain skill levels, it makes no sense to retire people there are no replacements possible.
- Example of UNZA retirements.
- A surplus of skills external to the organization may, however, justify retirements.
- In some developed countries, the shortage of skills is compensated by migration policies to attract foreign labor.

INVESTMENT DEMOGRAPHICS

- In corporate planning, investors rely on demographic information to decide where to invest in order to maximize the returns on investment.
- Investors look at population size and composition because these determine the opportunities for market size and opportunities for demand for their goods and services.
- A large population for example means a large market for their goods and services.
- A large young population presents opportunities for investments in goods and services required by younger people such as entertainment facilities, sports, fashions, etc.

INVESTMENT DEMOGRAPHICS

- **Investors will also rely on forecasting of future demographic trends using projections so that they know future demand for their goods and services so that they can decide whether to expand/increase their investments.**
- **In the United States a prominent business executive, Harold Levine, Vice President of E.F. Hutton, an investment company has suggested that in making an investment decision, 20 percent is based on demographics, 30 percent on the economic situation, and 50 percent on the individual characteristics of the individual company**

DOMESTIC DEMOGRAPHICS

- **This focuses on the demographic trends in the local market. That may influence the production of and demand for goods and services domestically.**

DEMOGRAPHIC TRENDS AND INVESTMENT PATTERNS

- **These trends include changes in the population size, composition, and distribution, migratory patterns (both internal and international), etc.**
- **Such changes may result in changes in the following:**
 - **The size of the market for goods and services which is influenced by the size and composition of the population.**
 - **Changes in the population with more purchasing power**
 - **Changes in the number of professional/managerial or middle class**
 - **Changes in expatriate population because of international migration**

INFLUENCE OF DEMOGRAPHIC TRENDS ON INVESTMENT PATTERNS

- **In turn, such demographic situations and trends may influence the location of industries.**
- **Domestic demographic situations and trends influences businesses in making decisions such as:**
- **Making appropriate site selections and identifying market areas for the establishment of retail centres like malls, factories, hotels, and other facilities, etc.;**
- **Forecasting demand and sales for goods and services on the basis of changes in population composition and distribution.**

OTHER INVESTMENT OPTIONS

- **Changes in the population composition have also been found to influence demand for alternative domestic investment options such as bonds, stocks, and treasury bills.**
- **As one investment company has observed “the attraction of bonds and stocks is partly a matter of demographics.**
- **Young and upwardly mobile people are more likely to invest in stocks and shares because they provide long – term investments.**
- **Older populations may invest in those that yield dividends in the short – term such as treasury bills.**
-

INTERNATIONAL DEMOGRAPHICS

- Investors often act in response to global demographic trends.
- Many companies look around the world for investment possibilities using global demographic trends as a yardstick.
- For example, the increase in investments by American companies in China is in response to the large market that such a large population represents.
- Investments of the Chinese in Zambia in the copper industry is a response of the huge population to raw materials from a developing country.
- Although some investors shy away from developing countries because of the risks inherent potential economic and political instability, others are willing to take a risk because of the huge demographic potential that may translate into enormous profits.

THE DEMOGRAPHIC DIVIDEND ADVANTAGE

- UNFPA stated that, “A country with both increasing numbers of young people and declining fertility has the potential to reap a demographic dividend. [\[3\]](#)
- Due to the dividend between young and old, many argue that there is a great potential for economic gains.
- In order for economic growth to occur the younger population must have access to quality education, adequate nutrition and health including access to sexual and reproductive health.

MARKETING DEMOGRAPHICS

- **Market research is the process of gathering, analyzing and interpreting**
 - **1. information about a market,**
 - **2. about a product or service to be offered for sale in that market,**
 - **3. and about the past, present and potential customers for the product or service;**
- **It also involves research into the characteristics, spending habits, location and needs of a business's target market, the industry as a whole, and the particular competitors faced.**
-
- **There are two important aspects of marketing namely segmentation and targeting involved in this process.**

SEGMENTATION

- **Segmentation refers to the manufacturing and packaging of products or the provision of services that appeal to specific socio demographically identifiable groups within the population.**
-
- **This means tailoring goods and services to specific groups.**
-
-
- **In other words, produce and package goods in such a way that they will be attractive to specific socio-demographic groups.**

SEGMENTATION

- **The packaging of some services in the Zambian market conforms to this dictum.**
- **Freshview /Ster – Kinekor – may appeal to the young**
- **Times – young and middle age**
- **O'Hagan - young and middle aged**
- **Boutiques – women**
- **Planet books/Bookworld Grey Matter**

TARGETING

- Targeting involves picking out particular socio-demographic characteristics of people who might purchase what you have to offer.
- This is done through market research in order to know something about the geographical location and socio-demographic characteristics of your actual and potential clientele, their tastes, preferences, and needs for goods and services
- Once this is done, then one appealing to the consumer tastes and behaviour reflected in those characteristics.
-
- This means examining in detail such socio- demographic and geographic characteristics such as:
-

TARGETING

- **Residence or community they stay in.**
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- **Age – People at different ages have different needs and tastes for products.**
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- **Radio Phoenix appeals to the young**
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- **Gender – Women represent a demographic category increasingly targeted by businesses, and advertisements.**
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- **Cosmetics to females**
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TARGETING

- **Education - This an important sociodemographic variable and a good predictor of consumer tastes.**
- **Wines to the Educated and high incomes**
- **Income – This represents purchasing power in terms of disposable income**
- **Ethnic groups, religious and racial groups provide identifiable targets for marketers.**

-

CLUSTER MARKETING

- This involves targeting geographical areas such as neighborhoods with a set of shared socio-demographic characteristics.
- It operates based on the adage that: “Birds of a feather flock together”.
- Neighborhoods can be identified on the basis of shared sociodemographic attributes, tastes and demands for goods and services.
- Life styles, tastes and demands for goods and services tend to coincide with sociodemographic characteristics of specific geographical areas.

CLUSTER MARKETING

- **This facilitates marketing to specific groups.**
- **EXAMPLES**
- **Establish Nandos in Kabulonga**
- **Tastes/and Demographic – Kabulonga or Chibuku Tavern in Bauleni.**

INSURANCE

- **Determination of life insurance premiums depends on demographic techniques and information.**
- **The widespread use of the life table in insurance companies is a reflection of this.**
- **Life expectancy is used to determine the insurance of human as well as non-human populations.**

PLANNING FOR SOCIAL SERVICES

- **CITY AND REGIONAL PLANNING**
- **Local agencies (City and municipal councils) use demographics for the adequate provision of services for their communities.**
- **Local area estimates and projections for the level of townships especially in the compliance with the policy of decentralization:**
 - **Where are the new roads to be built**
 - **How much water needs to be stored in reservoirs**
 - **Where new sewage pipes need to go**
 - **How many police officers and fire fighters must be recruited?**

PLANNING FOR SOCIAL SERVICES

- **EDUCATION**

- Demand for the provision of educational facilities is determined by demographic profile of the population.
- Schools recruitment and enrollment rise and fall on demographic processes such as fertility, mortality, and migration.
- Entry at primary level is directly dependent on fertility .
- Subsequent progression from primary to secondary levels are influenced by mortality, migration, as well as dropout rates.

PLANNING FOR SOCIAL SERVICES

- **University enrolment in turn relies on output from secondary school system.**
- **Quality of education is determined by demographic trends – are facilities commensurate with the number of students?**
- **Planning for future expansion of education system is also demographically determined.**

PLANNING FOR SOCIAL SERVICES

- **HEALTH**

- Demand for medical services and physicians is subject to demographic change.

- **GENDER**

- The size of the female population in the reproductive ages will determine demand for maternal wards and under five services

PLANNING FOR SOCIAL SERVICES

- **AGE**
- Older and younger people also exert different demands on health facilities.
- The greater the number of infants in the population the greater is the demand for pediatric services.
- The elderly utilize health services more frequently than younger people as they have greater medical needs.
- This may lead to greater demand for geriatric services.
- Younger people and the middle aged may rely more on self – care thus their demand for medical services may be more stable.

POLITICAL PLANNING

POLITICAL REPRESENTATION

- This is influenced by population changes and affects distribution of power.
 - Apportionment of districts constituency creation is influenced by population size and distribution.
- Larger units must have greater representation more constituencies.
- The electoral college system in the US operates on this principle.
- Proportional representation gives each political party a number of seats in parliament proportionate to the percentage of votes obtained.
- Attainment of city status is in part influenced by population size.

POLITICAL PLANNING

- **CAMPAIGN STRATEGY**

- Demographic information and techniques also crucial in campaign strategy.

- **THE DEMOGRAPHY OF THE VOTING**

- Demographic characteristics such as age, sex, ethnicity, and education are strongly related to the likelihood of voting and, to some extent to political preferences.
- In fact, effective campaign strategy must take into account demographic such as:
- Population size – how many voters are in a province, district, constituency:?
- Population composition – how many youths, middle aged, old people; men or women; college educated, not educated; married or single; black or white or Hispanic, tribe A or tribe B, C, D, E, F?

POLITICAL PLANNING

- **GEODEMOGRAPHICS**

- **This refers to the tendency of politicians appealing to potential voters in one's district with whom they have shared sociodemographic characteristics.**
- **The growing trend is to let the locals decide who their candidate is.**
- **On this basis, political planners therefore strategize “package and sell” a candidate like a product on sale – is he or she one of us?**
- **Geodemographics is an integral part of an election campaign strategy in most countries.**

POLITICAL PLANNING

- **OPINION POLLS**
- **These are used to determine voter preferences.**
- **Ideally, they should use scientific sampling techniques to gauge voter preferences.**

LEGISLATIVE ANALYSIS

- Demographic patterns and trends on influence legislative action.
- In turn, legislative action may influence demographic patterns and trends.
- FERTILITY
- Laws pertaining to marriage.
- Legislation to outlaw underage marriages may have long-term impact on fertility.
- Same sex marriages may also impact on fertility.
- Abortion laws may reduce fertility.
- Increase in population size due to high fertility may influence enactment of family planning laws.
- Progressive tax in the past gave a rebate for each additional child could have pushed fertility up.

- **MORTALITY**

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- Expansion of medical facilities may reduce mortality
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- National health insurance policy may reduce mortality.
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- In the US, Obamacare versus Trumpcare
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- **LEGISLATION ON RETIREMENT**

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- Changes in life expectancy can result it in legislative decision by parliament to change the age of retirement.
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- The aging of the population in US has led to social security policies meant to change taxation laws on the young and working population to sustain the social security scheme.
-
-

LEGISLATIVE ANALYSIS

- **MIGRATION**
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- **INTERNATIONAL MIGRATION**
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- **Laws on citizenship and naturalization**
- **Trump's executive order for the construction of a wall on the Mexican Border.**
-
- **INTERNAL MIGRATION**
-
- **Laws to regulate population movements**
-
- **SOUTH AFRICA**
-
- **Bantustan policy and influx laws**
-
- **ZAMBIA**
-
- **Poll and hut tax and the repatriation through the chibalo system.**

CAREERS IN DEMOGRAPHY

- **UNIVERSITIES:**
-
- **Research**
- **Teaching**
- **Staff Development Fellowship program**
-
- **GOVERNMENT/PUBLIC SECTOR**
-
- **Health – Biostatistics**
- **Census for planning**
- **Survey research**
- **Planning/decision making**
- **Agriculture**
- **Education**
- **Labour**
- **Registration of births/deaths**

CAREERS IN DEMOGRAPHY

- **CITY AND MUNICIPAL GOVERNMENT PLANNING**

- **Housing development**
- **Recreation facilities**
- **Markets/other facilities**

- **INTERNATIONAL ORGANIZATIONS**

- **UNFPA/UNICEF**

-

- **PRIVATE ORGANIZATION**

-

- **NGOs of world vision, ZARD**
- **Data collection and analysis**
- **Consultancy**

-

- **BUSINESS AND INDUSTRY**

- **Marketing**
- **Insurance**

BASIC TOOLS OF DEMOGRAPHY

- **COUNT**

- The absolute number of a population or any demographic event occurring in a specified area in a specified time period.
- An example is the Zambia's population of 13m.

- **RATES**

- The frequency of demographic events in a population in a specified time period.
- Rates tell how frequently an event is occurring or how common it is.
- Crude rate are rates computed for an entire population.
- Specific rates are rates computed for a specific subgroup – usually the population at risk of having the event occur.

BASIC TOOLS OF DEMOGRAPHY

- **RATIO**

- The relation of one population subgroup to another in the same population; that is, one subgroup divided by another.
- The number of men divided by the number of women to men in the population.

- **PROPORTION**

- The relation of a population subgroup to the entire population; that is, a population subgroup divided by the entire population.
- The number of people who can read and write divided by the total population gives the literacy

- **PERCENTAGE**

- This is a proportion multiplied by 100.

BASIC TOOLS OF DEMOGRAPHY

- **CONSTANT**

- An unchanging arbitrary number by which rates, ratios, or proportions can be multiplied to express these measures in more understandable fashion.
- The choice of a constant depends on how frequently an event occurs in the population.
- To explain rare events more meaningfully, use a large constant; for not so rare events use a smaller constant.

- **PROBABILITY**

- The likelihood of occurrence of an event.
- An example of this is the probability of dying – q_x value in a life table complemented by the probability of surviving - p_x .

MEASURES OF POPULATION SIZE

- **This is the total number of people in absolute terms within a defined territory or space.**
- **The population of a province, district, city, or township or country is often given in terms of an absolute count.**
- **Population size can be expressed in relative terms as a percentage of the total especially when making comparisons between different regions like provinces.**

MEASURES OF POPULATION SIZE

- **ZAMBIA 2000**

Central	1,012,257	10.2
Copperbelt	1,581,221	16.0
Eastern	1,306,173	13.2
Luapula	775,353	7.8
Lusaka	1,391,329	14.1
Northern	1,258,696	12.7
Northwestern	583,350	5.9
Southern	1,212,124	12.3
Western	765,088	7.7
Total	9,885,591	100.0

MEASURES OF POPULATION SIZE

- **ZAMBIA 2010**

Central	1,267,803	9.72
Copperbelt	1,958,623	15.01
Eastern	1,606,319	12.31
Luapula	958,976	7.35
Lusaka	2,198,996	16.86
Muchinga	743,109	5.70
Northern	1,117,903	8.57
Northwestern	706,462	5.41
Southern	1,606,793	12.32
Western	881,524	6.76
Total	13,046,508	100

MEASURES OF THE POPULATION COMPOSITION

- This refers to the constituent elements (parts, fractions or segments) that make up a population.
- This is the profile or complexion of the population.
- The major elements of a population include:
 - Age
 - Sex
 - Marital status
 - Education
 - Employment status

MEASURES OF THE POPULATION COMPOSITION

- In the analysis of population, one must be able to understand and measure some of the following concepts:
- **PERCENTAGE DISTRIBUTION**
- This refers to the proportion (or percentage) of the population in age groups, sex groups etc.
- A simple formula for this is:
- $(X/N) \times 100$
- Where X is the population in a particular age group or sex and N is the total population.
- According to the 2000 census, 51 percent of the population was female while 49 percent was male.
- In 2000, 45.3 percent of the population was under 15 years (i.e., between 0-14 years)

MEASURES OF THE POPULATION COMPOSITION

- **EXAMPLE 1**
- **ZAMBIA 2000**

SEX	POPULATION	PERCENT
MALE	4,591,731	49.2
FEMALE	4,745,678	50.8
TOTAL	9,337,409	100.0

MEASURES OF THE POPULATION COMPOSITION

- **EXAMPLE 2**
- **ZAMBIA 2000**

AGE GROUP	POPULATION	PERCENT
0-14	4,323,448	46.3
15-64	4,758,697	51.0
65+	255,280	2.7
TOTAL	9,337,425	100.0

MEASURES OF THE POPULATION COMPOSITION

- **SEX RATIO**
- This refers to the number of males per 100 females.
- Sex ratio = $(M/F) \times 100$

MEASURES OF THE POPULATION COMPOSITION

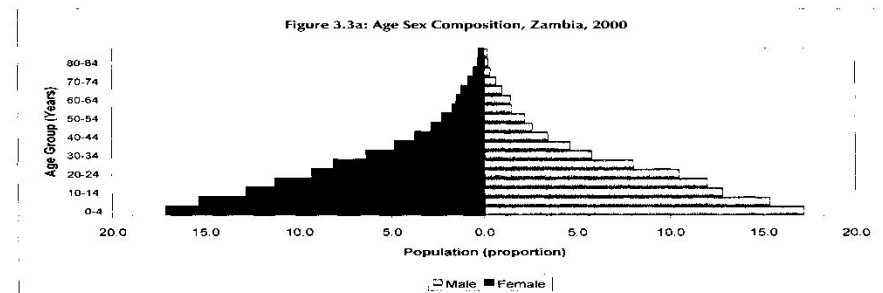
- **$SR = (4,591,731 / 4,745,678) * 100$**
- **$= 96.8$**
- **There were 97 men for every 100 women in Zambia in 2000.**

MEASURES OF THE POPULATION COMPOSITION

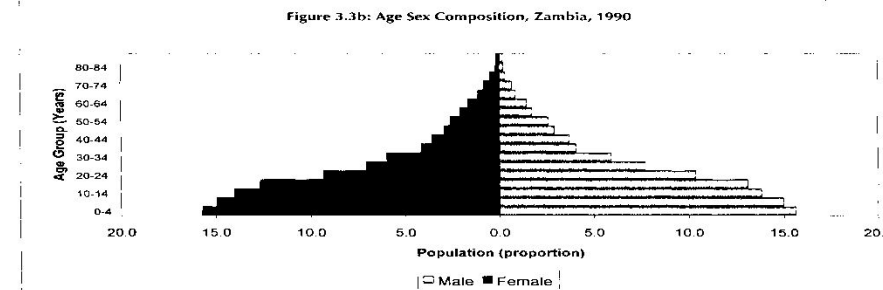
- **THE POPULATION PYRAMID**

This is a special type of bar graph representing the distribution of the population by age and sex.

MEASURES OF THE POPULATION COMPOSITION



Source: CSO, 2000 Census of Population and Housing



Source: CSO, 1990 Census of Population and Housing

MEASURES OF THE POPULATION COMPOSITION

- **THE DEPENDENCY RATIO**

-
- This refers to the ratio of the young (those aged under 15 years) and the old (those aged above 65 years) to economically active population (those aged between 15 and 64 years).
- $DR = (\text{Population } <15 + \text{population } >65 / \text{Population } 15-64) * 100$
- $DR(2000) = (4,578,728 / 4,758,697) * 100$
- $= 96.2$
- In 2000, there were 96 dependents per 100 economically active persons in the population.

MEASURES OF THE POPULATION COMPOSITION

- **CHILD DEPENDENCY RATIO**

- $CDR(2000) = (4,323,448)/4,758,697 * 100$

- $= 90.9$

- In 2000, there were 91 child dependents per 100 economically active persons in the population.

- **AGED DEPENDENCY RATIO**

- $ADR(2000) = (255,280)/4,758,697 * 100$

- $= 5.4$

- In 2000, there were 5 aged dependents per 100 economically active persons in the population.

MEASURES OF THE POPULATION COMPOSITION

- **THE MEDIAN AGE**

- This is the age which divides the population into two equal parts.
- It is often used as a measure of aging of population.
- Populations with medians under 20 may be considered as "young";
- Populations with medians between 20 to 29 as "intermediate"; and
- Populations with medians 30 or over as "old".
- The median age according to the 2000 Census was 17.7 years.
- Half the population in Zambia is below 18 years.
- This puts Zambia in the young category.

MEASURES OF THE POPULATION COMPOSITION

- **AGE CLASSIFICATIONS**

-
- **Childhood: Infancy (age 0) to about age 9.**
- **Adolescence: 10-19 years, with the following subdivisions:**
 -
 - **Early adolescence: 10 – 14 years**
 - **Late adolescence: 15 – 19 years**
-
- **Youth**
-
- **15 – 34 years (WHO)**
-
- **Early youth: 15 – 24 years (UNICEF)**
- **Mature youth: 25 – 34 years**
-

MEASURES OF THE POPULATION COMPOSITION

- **Middle age: 35 – 64**
- **There are variations to this.**
- **Some put this between 40 – 65 years (others, this between 35 and 65 years; others, between 40 and 60 years; and in some cases even 25 – 75 years).**
- **Prominent psychologist, Erik Erikson, sees it ending a little later and defines middle adulthood as between 40 and 65.**

This is a period in one's life, according to one definition, "when you are no longer young, but not yet old".

MEASURES OF THE POPULATION COMPOSITION

- **Old age:**
-
- **Over 65 years (ILO/UN)**
-
- **(This is the reason incidentally why most Western society use age 65 as retirement age)**
-
- **Some argue that delineating sub-groups in the 65+ population enables a more accurate portrayal of significant life changes.**
-
- **Young old :65-74) years**
-
- **Old : 74-84 years**
-
- **Very old: 85+**

MEASURES OF THE POPULATION COMPOSITION

- **NEW PERSPECTIVES FROM WHO (2017)**
- **World Health Organization (WHO) had done a new research recently, according to average health quality and life expectancy, and defined a new criterion that divides human age as follows:**
 - **0-17 years old: underage**
 - **18-65 years old: youth/young people**
 - **66-79 years old: middle-aged**
 - **80-99 years old: elderly/senior**
 - **100+ years old: long-lived elderly**

MEASURES OF POPULATION DISTRIBUTION

- This refers to the location of the population in space or the settlement patterns and dispersal of the population within specific areas.
- Population can be distributed spatially within provinces and districts or by rural and urban residence.

MEASURES OF POPULATION DISTRIBUTION

- **RURAL URBAN POPULATION IN ZAMBIA (2000)**

Rural	5,990,356	60.6
Urban	3,347,069	33.9
Total	9,337,425	94.5

MEASURES OF POPULATION DISTRIBUTION

- This show that thirty eight percent of the population in Zambia is urban whilst sixty two percent is rural.
- Residence in a given space determines population density

MEASURES OF POPULATION DISTRIBUTION

- **Population distribution can also be expressed in terms of population density.**
- **Population density is expressed as the number of people per unit of land area.**
- **Density is reported in terms of number of people per square kilometre.**
- **PD = Population/Area**

MEASURES OF POPULATION DISTRIBUTION

- POPULATION DENSITY BY PROVINCE 2010

PROVINCE	DENSITY
Lusaka	100.4
Copperbelt	62.5
Eastern	24.6
Luapula	19.0
Southern	18.8
Central	13.4
Northern	11.9
Western	7.0
North-western	5.6

MEASURES OF POPULATION DISTRIBUTION

- TEN MOST DENSELY POPULATED TOWNS AND CITIES IN ZAMBIA 2010

Lusaka	4,841.6
Kitwe	671.9
Ndola	412.7
Livingstone	204.4
Luanshya	188.8
Kalulushi	132.7
Kabwe	129.1
Chingola	125.2
Mufulira	98.7
Chililabombwe	88.2

MEASURES OF POPULATION DISTRIBUTION

- TEN LEAST DENSELY POPULATED TOWNS AND CITIES IN ZAMBIA 2010

Mufumbwe	2.7
Kasempa	3.2
Sesheke	3.2
Itezhi-tezhi	4.0
Mpika	5.2
Lukulu	5.2
Chama	5.8
Kazungula	5.8
Shang'ombo	5.9
Zambezi	6.3

MEASURES OF DEMOGRAPHIC VARIABLES

- **The major demographic variables demographers are interested in are:**
 - **Fertility**
 - **Mortality**
 - **Migration**
- **The interaction of these three variables is important in understanding population change.**

CONCEPTS AND MEASURES ASSOCIATED WITH FERTILITY

- **Fertility refers to the actual level of reproductive performance in a population based on the number of live births that occur.**
- **Fertility can also be viewed as the number of live births in a population.**
- **A live birth occurs when a fetus, whatever its gestational age, exits the maternal body and subsequently shows any sign of life, such as voluntary movement, heartbeat, or pulsation of the umbilical cord, for however brief a time and regardless of whether the umbilical cord or placenta are intact.**
- **Fertility differs from fecundity which is the physiological capability of couples to reproduce.**

CONCEPTS AND MEASURES ASSOCIATED WITH FERTILITY

- **Fertility is affected by the following factors:**
- **Fecundity**
- **Age at marriage or cohabitation**
- **The availability and use of family planning**
- **Economic development**
- **Status of women**
- **The age-sex structure**

CRUDE BIRTH RATE

- This is basic measure of fertility referring to the number of births per 1000 population in a given year.
- The formula used is:
 - $\frac{\text{Number of births}}{\text{Mid - year population}} \times 1000$
- India (1983)
 - $\text{CBR} = (25,745,000 / 735,596,000) * 1000$
 - = 35
- In India in 1983, there were 35 live births per 1,000 population.
- In Zambia, in 2000, the CBR was estimated to be 36 live births per 1000 population.

USE OF THE MID-YEAR POPULATION

	JANUARY 2020	DECEMBER 2020	JUNE 2020
BIRTHS	650,000	650,000	650,000
POPULATION	18,000,000	18,500,000	18,250,000
	0.036	0.035	0.036
CBR	36.1	35.1	35.6

CHILD – WOMAN RATIO

- This is the ratio of all children aged 0-4 to women aged 15-49 in a population.
- $\frac{\text{Children aged 0-4}}{\text{Women aged 15-49}} \times 1000$

Zambia 2000

$$\begin{aligned}\text{CWR} &= (1,656,720 / 2,210,008) * 1,000 \\ &= 750\end{aligned}$$

- For every 1,000 women in reproductive ages (15-49) there 750 children in age group 0-4.

GENERAL FERTILITY RATE

- This is a more refined measured of fertility as it attempts to relate the number of births to women in their reproductive ages.
- $\frac{\text{The number of births}}{\text{The number of women aged 15-49}} \times 1000$
- Puerto Rico (1980)
- $GFR = (73,060 / 826,690) * 1,000$
- = 88.4
- There were 88 births per 1,000 women aged 15-49 years in Puerto Rico in 1980.
- For Zambia, the general fertility rate in 2000 was 152.

PARITY AND COMPLETED FAMILY SIZE

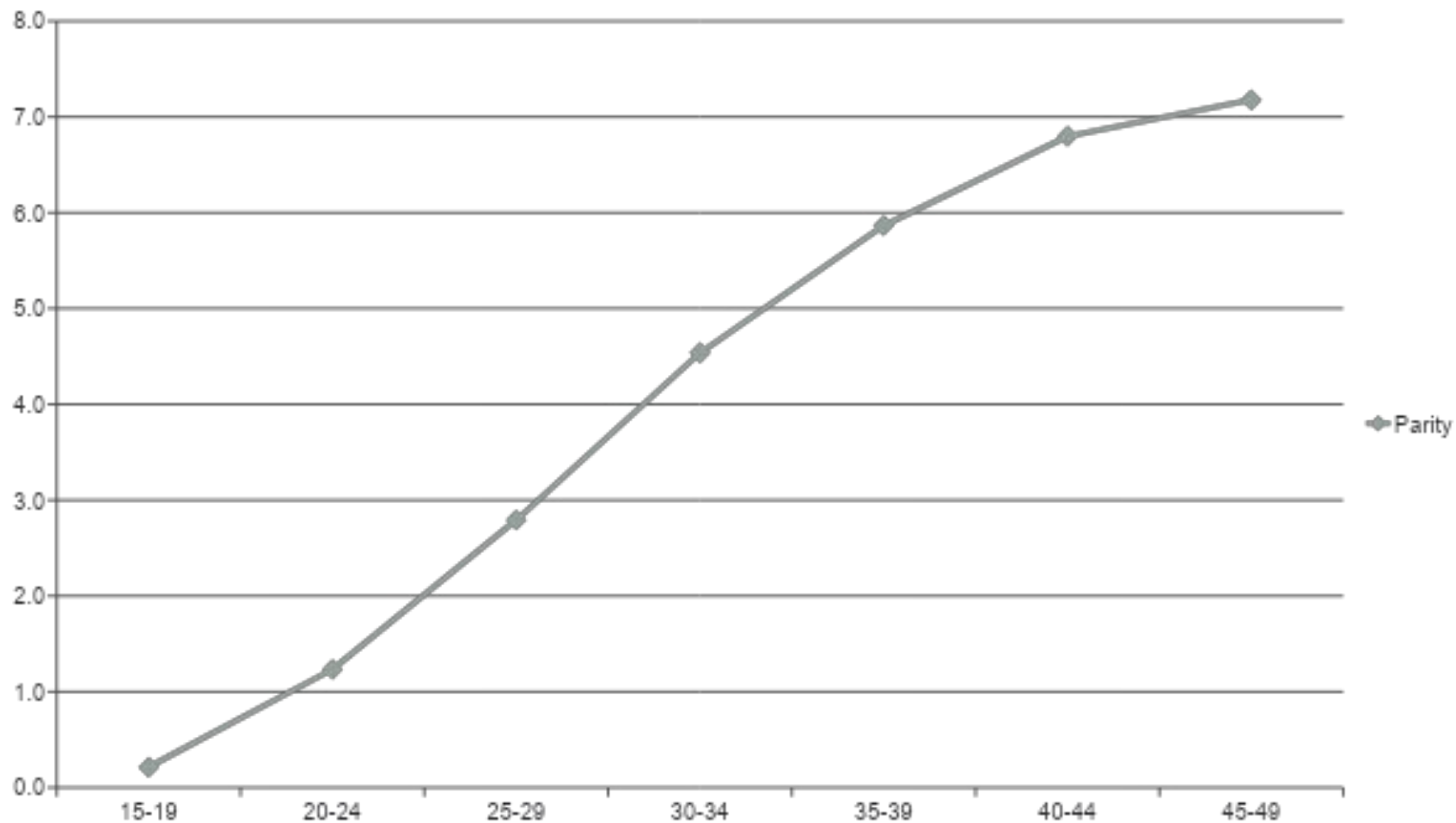
- **PARITY**
- **This refers to the average number of children born alive to women in a particular age group.**
- **COMPLETED FAMILY SIZE OR MEAN PARITY**
- **The "number of children ever born" at various ages provides one measure of a population's fertility.**
- **This is the number of children ever born to women 45-49 who have completed reproduction.**
- **Completed family size was 6.8 in 2000; in 1990, it was 7.2.**

PARITY AND COMPLETED FAMILY SIZE

- **COMPLETED FAMILY SIZE AND PARITY: ZAMBIA (1990)**

Age group	Women	Children ever born	Parity
15-19	47,422	10,099	0.2
20-24	36,981	45,618	1.2
25-29	27,406	76,524	2.8
30-34	20,925	94,942	4.5
35-39	13,855	81,276	5.9
40-44	12,862	87,443	6.8
45-49	10,547	75,698	7.2

PARITY AND COMPLETED FAMILY SIZE



AGE SPECIFIC FERTILITY RATE

- **This is the fertility measured at each age or age group for women in the reproductive age group between 15 and 49 years.**
- **Fertility rates can also be obtained for specific age groups for comparison over time or to see differences in fertility behaviour at different ages.**

AGE SPECIFIC FERTILITY RATE

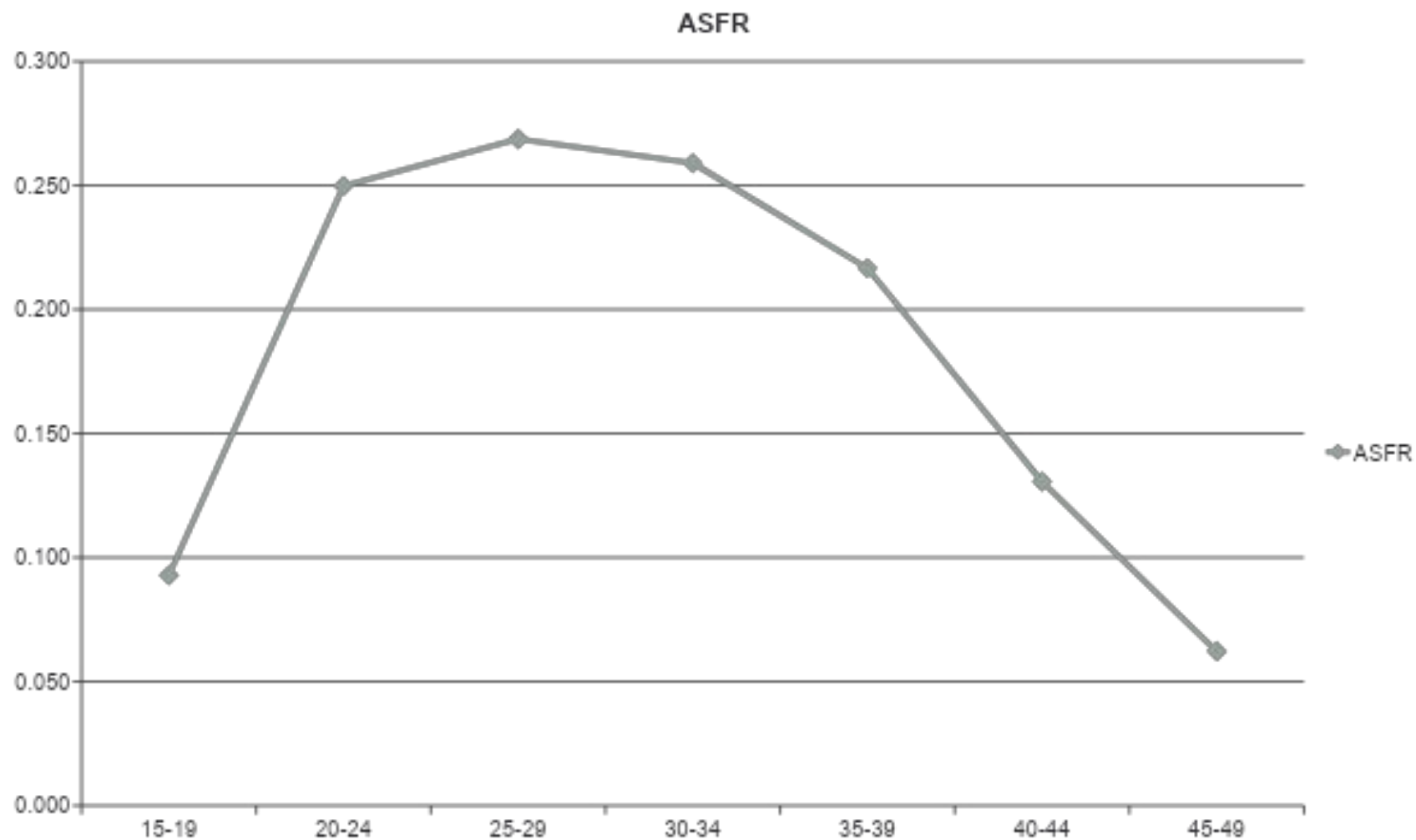
- For example, the age specific fertility rate for women in the age group 20-24 is simply,
- $\frac{\text{Number of births to women aged 20-24}}{\text{Number of women aged 20-24}} \times 1000$
- Zambia (1990)
- $\text{ASFR (20-24)} = (9,238/36,981)$
- $= 0.250$
- If we apply a constant of 1,000, this means that for every 1,000 women in the age group 20-24, 250 births took place.

AGE SPECIFIC FERTILITY RATE

- **AGE SPECIFIC FERTILITY RATE: ZAMBIA (1990)**

Age group	Women	Births last 12 months	Fertility
15-19	47,422	4,407	0.093
20-24	36,981	9,238	0.250
25-29	27,406	7,364	0.269
30-34	20,925	5,420	0.259
35-39	13,855	3,002	0.217
40-44	12,862	1,680	0.131
45-49	10,547	656	0.062

AGE SPECIFIC FERTILITY RATE



TOTAL FERTILITY RATE

- The total fertility rate (TFR) is the average number of children that would be born alive to a woman (or group of women) during her lifetime if she were to pass through all her childbearing years conforming to the age-specific rates of a given year.
- The TFR sums up in single number the fertility of all women at a given point in time.
- In effect, it says: This is the total number of children a woman would have at age 15-19 as all women that age in given year; at age 20-24 as all women that age in given year ... and so on up to age 49.
- TFR in 1990 was:
 - $\text{Sum} = 1.280 \times 5$
 - $= 6.4$
- In 1990, TFR was 6

GROSS REPRODUCTION RATE

- The gross reproduction rate (GRR) is the average number of daughters that would be born to a woman (or group of women) if she passed through her childbearing years conforming to the age specific fertility rates of a given year.
- This rate is like the TFR except it counts only daughters and literally measures "reproduction" - a woman reproducing herself by having a daughter.

GROSS REPRODUCTION RATE

- Calculation of GRR involves multiplying the TFR by the proportion of females in the population.
- In 1990, GRR was 3.3
- This means a woman would reproduce herself three times
- GRR declined to 2.3 in 2000.
- This means a woman would reproduce herself two times.

NET REPRODUCTION RATE

- The net reproduction rate (NRR) is the average number of daughters that would be born to a woman (or a group of women) if she passed through her lifetime from birth conforming to the age-specific fertility and mortality rates of a given year.
- This rate is similar to the GRR, but is always lower than the GRR because it takes into account the fact that some women will die before completing their childbearing years.

NET REPRODUCTION RATE

- In Zambia, the GRR in 1990 was 3.3 while the NRR was about 2.
- This means that if the 1990 fertility levels were to continue, a woman in Zambia would produce 3 daughters on average during her lifetime.
- Since the NRR is 2, this means only two daughters would survive through the childbearing years.
- NRR in 2000 was 1.7

REPLACEMENT LEVEL FERTILITY

- **This is the level of fertility at which a cohort of women, on average, have only enough daughters to replace themselves in a population.**
-
- **An NRR of 1.00 is equal to replacement level.**
-

REPLACEMENT LEVEL FERTILITY

- Once replacement level fertility has been reached, births will gradually reach equilibrium with deaths, and in the absence of immigration and emigration, a population will ultimately stop growing, and become stationary.
-
- A population which has achieved replacement level or below replacement level fertility may still continue to grow for some decades, since past high fertility may have led to a high concentration of women in childbearing ages and hence total births continue to exceed total deaths.
-
- Finland with NNR of 0.83 has below replacement level fertility.
-
- Thus it may take two or three generations (50 to 70 years) before each new birth is offset by a death in a population.
-

CONCEPTS AND MEASURES ASSOCIATED WITH FERTILITY

- **POPULATION MOMENTUM**

-
- The tendency for a population to grow even after replacement fertility has been achieved is known as population momentum.
-

- **CONTRACEPTIVE PREVALENCE RATE**

-
- This is the percentage of women currently using specific family planning methods.

- **UNMET NEED FOR FAMILY PLANNING**

- Percentage of women either do not want children want to wait for two or more years before having another child but are not using contraception.
- Women with unmet need and met need together constitute total demand for family planning.
- 2001 – 2002 unmet need
- 27 percent of married women had unmet need



CONCEPTS AND MEASURES ASSOCIATED WITH MORTALITY

- **This refers to the effect of death on the population.**
- **Mortality also refers to deaths as a component of population change.**
- **The rate at which mortality occurs is linked to many factors such as age, sex , race, occupation, , social class, education, etc**
- **The incidence of death in the population can reveal much about a population's standard of living and health care.**

CRUDE DEATH RATE

- The crude death rate (CDR) is conceptually similar to crude birth rate.
- It refers to the number of deaths per 1000 population in a given year. The formula used is:
- $\frac{\text{Number of deaths}}{\text{Total population}} \times 1000$

Israel (1983)

- $\text{CDR} = (27,818 / 4,037,620) \times 1,000$
- $= 6.9$
- In Israel, there were 7 deaths per 1,000 of the population. In 1983
- In 1990, the CDR in Zambia was 19.6, i.e., there were 19.6 deaths per 1,000 population in Zambia.
-

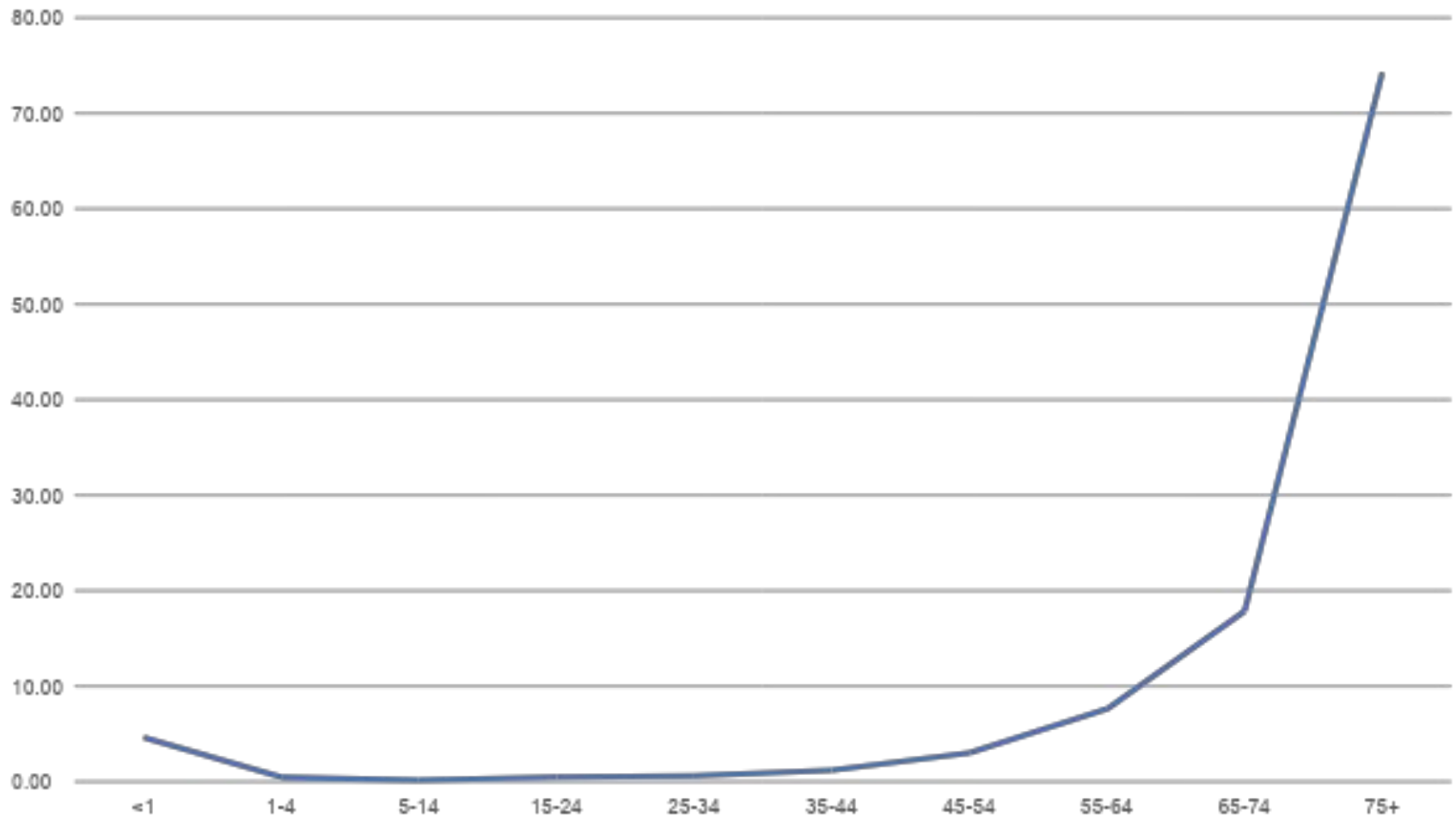
AGE SPECIFIC DEATH RATE

- Death rates can be obtained for specific age groups for comparison of mortality at different ages or a change in mortality at the same age over time.
- Comparisons can also be made between countries or areas.
- Since mortality varies greatly by sex, age specific rates are often given separately for males and females.
- Deaths of people aged (15-24) x 1000
- Total population aged (15-24)
-
- $ASDR = (2,084 / 1,267,400) * 1,000$
- = 1.6
- There about 2 deaths for every 1,000 persons in the age group 15-24.

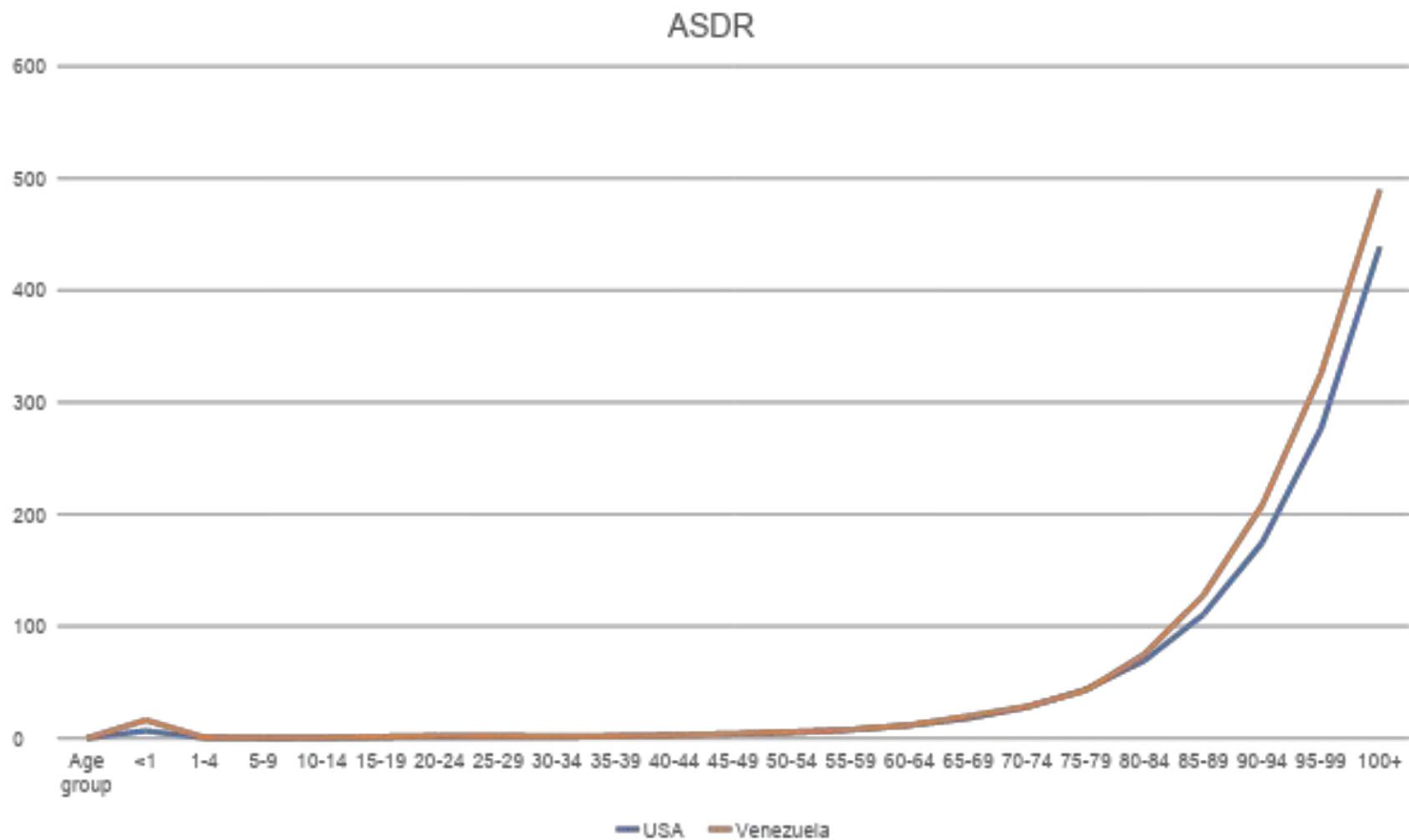
AGE SPECIFIC DEATH RATE

Japan 1990			
Age group	Population (thousands)	Deaths	ASDR
<1	1,217	5,616	4.61
1-4	5,293	2,367	0.45
5-14	16,037	2,619	0.16
15-24	18,856	9,148	0.49
25-34	15,901	9,315	0.59
35-44	19,714	23,862	1.21
45-54	17,151	51,986	3.03
55-64	14,508	110,269	7.60
65-74	8,946	159,744	17.86
75+	5,989	444,930	74.29

CONCEPTS AND MEASURES ASSOCIATED WITH MORTALITY



AGE SPECIFIC DEATH RATE



INFANT MORTALITY RATE

- The infant mortality rate is the number of deaths to infants under 1 year of age per 1000 live births in a given year.
- Number of deaths to infants under age 1 x 1000
- Total live births
- Sweden 1984
- $IMR = (594/93,792) * 1,000$
- =6.3
- There were 6 deaths to infants under age 1 per 1,000 live births in Sweden in 1984.
- According to the 2000 census , there were 110 deaths to infants per 1000 live births in Zambia .

MATERNAL MORTALITY RATE

- The maternal mortality rate is the number of women who die as result of childbearing in a given year per 100,000 births in that year. Maternal deaths are those caused by complications of pregnancy and childbirth.
- $\frac{\text{Number of maternal deaths}}{\text{Total live births}} \times 100,000$
- US (1983)
- $\text{MMR} = (290/3,614,000) \times 100,000$
- = 8.0
- There were 8 maternal deaths per 100,000 live births in the US in 1983.
- According to the 1996 ZDHS, there were 649 maternal deaths per 100,000 live births in Zambia.
- The ZDHS 2007 maternal mortality rate for Zambia was 591 per 100,000 live births.

LIFE EXPECTANCY

- **This is an estimate of the average number of additional years a person can expect to live if the current age specific death rates continued.**
- **Although life expectancy can be calculated at each age, the most commonly cited life expectancy is the one at birth.**

LIFE EXPECTANCY

- Life expectancy is derived from a life table.
- A life table (also called a mortality table or actuarial table) is a table which shows, for each age, what the probability is that a person of that age will die before his or her next birthday.
- A life table can be used in the determination of the following:
 - The probability of surviving any particular year of age
 - The remaining life expectancy for people at different ages

LIFE TABLE

Age	<i>l</i>x	<i>n</i>qx	<i>n</i>p_x	<i>n</i>d_x	<i>n</i>L_x	<i>T</i>x	<i>e</i>x
0	100,000	0.00825	0.99175	825	99,257	7,757,864	77.6
1	99,175	0.00163	0.99837	162	396,311	7,658,607	77.2
5	99,013	0.00091	0.99909	90	494,840	7,262,296	73.4
10	98,923	0.00094	0.99906	93	494,382	6,767,456	68.4
15	98,830	0.00141	0.99859	139	493,801	6,273,074	63.5
20	98,691	0.00153	0.99847	151	493,075	5,779,273	58.6
25	98,540	0.00182	0.99818	179	492,249	5,286,198	53.7
30	98,360	0.00283	0.99717	278	491,105	4,793,948	48.7
35	98,082	0.00441	0.99559	433	489,328	4,302,843	43.9
40	97,649	0.00720	0.99280	703	486,489	3,813,515	39.1
45	96,946	0.01235	0.98765	1,197	481,738	3,327,026	34.3
50	95,749	0.02083	0.97917	1,994	473,759	2,845,288	29.7
55	93,755	0.03456	0.96544	3,240	460,672	2,371,529	25.3
60	90,514	0.05851	0.94149	5,296	439,332	1,910,857	21.1
65	85,218	0.08731	0.91269	7,440	407,491	1,471,526	17.3
70	77,778	0.13919	0.86081	10,826	361,825	1,064,035	13.7
75	66,952	0.22099	0.77901	14,796	297,771	702,210	10.5
80	52,156	0.35237	0.64763	18,378	214,835	404,439	7.8
85	33,778	1	0	33,778	189,604	189,604	5.6

LIFE TABLE

<i>Age</i>	<i>ex</i>
0	77.6
1	77.2
5	73.4
10	68.4
15	63.5
20	58.6
25	53.7
30	48.7
35	43.9
40	39.1
45	34.3
50	29.7
55	25.3
60	21.1
65	17.3
70	13.7
75	10.5
80	7.8

CONCEPTS AND MEASURES ASSOCIATED WITH MORBIDITY

- **MORBIDITY**
- **This refers to disease and illness in a population.**
- **It also refers to the incidence of disease and illness in a population.**
- **Demographers study morbidity because the outbreak of disease (or illness) impacts on mortality.**
- **Data about the frequency and distribution of a disease can aid in controlling its spread and in some cases lead to identification of its cause.**

INCIDENCE RATE

- This is the number of persons contracting a disease during a given time period per 1,000 population at risk.
- Other constants such as per 100,000 may also be used depending on the occurrence level.

- $IR = \frac{\text{Number of persons Developing a disease During a given time period}}{\text{Total population at risk}} \times 100,000$

Canada (1985)

- $= (2,280 / 25,358,500) * 100,000$
 - =9.0
- The incidence of German measles in Canada in 1985 was 9 persons developing the disease per 100,000 population.

PREVALENCE RATE

- This is the number of persons having a particular disease at a given point in time per 1,000 population. The rate includes all previously existing cases or well as new cases developing during the specified period .
- The prevalence rate is a “SNAPSHOT” of an existing health situation.
- It describes the health status of a population.
-
- $$\text{PR} = \frac{\text{Number of persons with a specific disease}}{\text{Total population}} \times 1,000$$
- $$= \frac{(4,740,000)}{233,981,000} \times 1,000$$
- $$= 20.3$$
- The prevalence of coronary heart disease in the US in 1983 was 20 persons per 1,000 population.
- The HIV prevalence rate in Zambia according to ZDHS 2007 was 14.3%

MIGRATION

- **This refers to the permanent movement of persons or groups over a significant distance.**
- **It often involves permanent change of address.**

WHAT MIGRATION IS

- **Internal migration – change of residence across boundaries of administrative areas within a country.**
- **Internal migrants are classified into out-migrants and in-migrants.**
- **International migration – change of residence involving the crossing of international boundaries**
- **International migrants are often classified in terms of immigrants and emigrants.**

DEFINITIONS AND CONCEPTS

- **AREA OF ORIGIN(DEPARTURE)**
- **The area from which a migrant moves.**
- **AREA OF DESTINATION (ARRIVAL)**
- **The area to which a migrant moves.**

INTERNAL MIGRATION

- **One could, for instance, move from Lusaka (Origin) to Kitwe (Destination).**
- **Those who move from Lusaka to Kitwe are out-migrants.**
- **In Kitwe, these are considered in-migrants.**

INTERNATIONAL MIGRATION

- **One could, for instance, move from Zambia to (Origin) Namibia (Destination) .**
- **Those who move from Zambia to Namibia are emigrants.**
- **In Namibia, these are considered immigrants.**

WHAT MIGRATION IS

- **Certain categories of the population are not included from this definition of migration.**
- **These include the following:**
 - **Nomadic populations**
 - **Seasonal workers**
 - **Persons with more than one residence**
 - **Tourists**
 - **Commuters.**

TYPES OF INTERNAL MIGRATION

- **RURAL – RURAL MIGRATION**

- This involves movement from one rural area to another, e.g., Shangombo to Kaputa

- **RURAL – URBAN MIGRATION**

- This involves movement from rural areas to urban areas, e.g., Kaputa to Kitwe

- **URBAN – RURAL MIGRATION**

- This involves movement from urban to rural area, e.g., Kitwe to Gwembe Valley.

- **URBAN – URBAN MIGRATION**

- This involves movement from urban to urban areas, e.g., Kitwe to Lusaka

CAUSES OF MIGRATION

	PUSH FACTORS: Negative factors in area of origin	PULL: FACTORS: Positive factors in area of destination
ENVIRONMENTAL	Poor Climatic conditions; droughts; famines; poor soil fertility; land shortages; pressure on land	Attractive climatic conditions; adequate arable fertile soils.
SOCIAL	High crime rates; absence social amenities; cultural differences; ethnic tensions	Low crime rates; “bright lights” of the city; presence of social amenities; cultural similarities; ethnic harmony; similarities in language
POLITICAL	Political repression and instability; dictatorship; one-party system.	Political freedom; democratic institutions; multiparty democracy.
ECONOMIC	Low levels of economic development; absence of economic opportunity; high unemployment levels; low incomes; poor working conditions.	High levels of economic development; economic opportunities; greener pastures; employment opportunities; business

MEASUREMENT OF INTERNAL MIGRATION

- The examples used here are based on data for selected districts in Gabon using place of birth and place of enumeration between 1960 and 1961

IN – MIGRATION RATE

- **EXAMPLE**
- $\frac{\text{Number of in-migrants}}{\text{Total population in area}} \times 100$
- In - migrants = 1,491
- Native or local population = 76,100
- $IM = (1,491/76,100) \times 100$
- = 2.0
- Interpretation: For every 100 local or native residents of the area, 2 newcomers moved in.

MEASUREMENT OF INTERNAL MIGRATION

- **OUT-MIGRATION RATE**

- Number of out-migrants x 100

- Total population in area

- Out - migrants=Number of people in area who are now living outside the area.

- Total population= Native or local population enumerated in area.

MEASUREMENT OF INTERNAL MIGRATION

- **EXAMPLE**

- **Out- migrants=5,386**

- **Native or local population = 76,100**

- **$OM = (5,386/76,100)*100$**

= 7.1

- **Interpretation: For every 100 local or native residents of the area , 7 left the area.**

MEASUREMENT OF INTERNAL MIGRATION

- NET MIGRATION
- This is the difference between people coming into an area minus those leaving an area.
- $NM = \frac{\text{Number of in-migrants} - \text{Number of out-migrants}}{\text{Native or local population}} \times 100$
- $= (1,491 - 5,386) / 76,100 \times 100$
- $= -5.1$
- Interpretation: The area experienced a net decrease of 5 people per 100 population through migration
- In net migration a negative means a loss in numbers; a positive value, a gain.

INTERNATIONAL MIGRATION – CONCEPTS AND DEFINITIONS

- **International migration represents the permanent or long – term movement across internationally defined boundaries.**
- **IMMIGRANT**
- **Immigration is moving into a country; an immigrant is one who arrives.**
- **A migrant to the area from a place outside the country.**
- **An “immigrant” is also referred to as a MOVER FROM ABROAD from outside the country into the country.**

INTERNATIONAL MIGRATION – CONCEPTS AND DEFINITIONS

- **EMIGRANT**

- **Emigration is moving out of a country;**
- **A migrant from the area to a place in another country.**
- **Emigration is moving out of a country. An émigré is one who has left the country.**

INTERNATIONAL MIGRATION – CONCEPTS AND DEFINITIONS

- **INTERNATIONAL MIGRANT**

- **This is an emigrant or immigrant or any a person who changes his or her country of abode.**
- **The UN ‘s definition of abode:**
- **“A person’s abode is the country where that person spends most of his/ her daily night rest over a period of a year.”**

STAGES IN IMMIGRANT STATUS

- **EXPATRIATE STATUS**

- An immigrant may start as a temporary migrant.
- Upon arrival in the country of destination, he is given expatriate status.

- **PERMANENT RESIDENCE**

- After some years of residence, he/she may later change status to permanent migrant or resident.

STAGES IN IMMIGRANT STATUS

- **NATURALIZED CITIZENSHIP**

- Depending on the prevailing laws of the country, the immigrant may apply for citizenship and become a naturalized citizen.

- **INTERGENERATIONAL IMMIGRATION STATUS**

- First generation immigrants are those who have made a permanent movement to another country from their country of origin.
- Second generation immigrants comprise their offspring.
- Third, fourth and later generations may result from future offspring.

INTERNATIONAL MIGRATION

- **Children of citizens of Zambia 6. (1) Any person who- Persons entitled to apply to be registered as citizens (a) has attained the age of twenty-one years; and (b) has been ordinarily resident in Zambia for a continuous period of not less than ten years immediately preceding that person's application for registration; shall be entitled to apply to the Citizenship Board, in such manner as may be prescribed by or under an Act of Parliament, to be registered as a citizen of Zambia.**

INTERNATIONAL MIGRATION

- **Refugees represent a special class of immigrants who are admitted under a special dispensation of the host country because of political persecution in the home country.**
- **Illegal immigrants are also called illegal aliens or undocumented immigrants who enter a country illegally in search of economic opportunity.**
- **In the US, overstayers are those who enter the country legally as students or tourists but decide to stay on.**

MAJOR TYPES OF INTERNATIONAL MIGRATION

- These include the following:
- Temporary movements
- Permanent movements
- Movements of individuals and families
- Movements of whole nations or tribes
- Movements of citizens or aliens
- Voluntary or forced movements
- Movements of civilians or military personnel
- Movements for work
- Movements for study or other purposes

MEASUREMENT OF INTERNATIONAL MIGRATION

- Measures of international migration data are often based on census or border and airport data.
- THE IMMIGRATION RATE
- This is the number of immigrants arriving at a destination per 1,000 population at that destination in a given year.
- Number of of immigrants * k
- Total population at destination

MEASUREMENT OF INTERNATIONAL MIGRATION

- **EXAMPLE OF SWEDEN**

- **IMMIGRATION RATE**

- Number of immigrants – people moving into Sweden in 1984 as recorded at points of entry= 31,486
- The total population of Sweden in 1984= 8,336,597
- Immigration rate = $\frac{31,486}{8,336,597} \times 1,000$
- $= 3.8$
- Interpretation: in 1984, the Swedish immigration rate was 3.8 immigrants new immigrants for every 1,000 residents.

MEASUREMENT OF INTERNATIONAL MIGRATION

- **THE EMIGRATION RATE**

- This is the number of emigrants departing an area of origin per 1,000 population at that area of origin in a given year.

$$\frac{\text{Number of emigrants}}{\text{Total population at area of origin}} * k$$

MEASUREMENT OF INTERNATIONAL MIGRATION

- **EXAMPLE OF SWEDEN**

- **Number of emigrants – people moving out of Sweden in 1984 as recorded at points of exit= 22,825**
- **The total population of Sweden in 1984= 8,336,597**
- **Emigration rate = $\frac{22,825}{8,336,597} \times 1,000$**
 - **8,336,597**
 - **= 2.7**
- **Interpretation: In 1984, the Swedish emigration rate was 2.7 emigrants for every 1,000 residents.**

MEASUREMENT OF INTERNATIONAL MIGRATION

- THE NET MIGRATION RATE
- This shows the net effect of immigration and emigration on an area's population.
- It is expressed as increase or decrease per 1,000 population of the area in a given year.
- $$\frac{\text{Number of immigrants} - \text{Number of emigrants}}{\text{Total population of area}} \times k$$
- $$\frac{31,486 - 22,825}{8,336,597} \times 1,000$$
- $$= +1.0$$
- Interpretation: In 1984, Sweden experienced a net increase of 1.0 persons per 1,000 population through migration.

IMPACT OF MIGRATION

	AREA OF ORIGIN		AREA OF DESTINATION	
	POSITIVE	NEGATIVE	POSITIVE	NEGATIVE
SOCIAL		Poverty; low living standards	Urbanism; melting pot	Crime; prostitution; street children phenomenon; uncontrolled vending
ECONOMIC	Remittances may boost rural economies	Reduced productivity; loss of productive workers; brain drain;	Brain gain; Expansion of the informal sector by enterprising migrants resulting in employment creation;	Pressure on resources; unemployment; low wage rates because of increase in labour supply

IMPACT OF MIGRATION

	AREA OF ORIGIN		AREA OF DESTINATION	
	POSITIVE	NEGATIVE	POSITIVE	NEGATIVE
ENVIRONMENTAL				High population density; unplanned settlements; lack of access to clean water supply
TECHNOLOGICAL	Transfer of skills			
DEMOGRAPHIC	Low sex ratios; old population; depopulation; female headed households			

POPULATION CHANGE

- **THE BALANCING EQUATION**
- **This process of population change is often depicted through the balancing equation.**
- **This presents population change as a function of the initial population and the interaction of births, deaths, and migration in the intervening period.**
-

POPULATION CHANGE

- $P_t = P_o + B - D + I - O,$
-
- where P_t = Current population
- P_o = Previous population
- D = Deaths between current and previous population
- B = Births between current and previous population
- I = In-migrants between current and previous population
- O = Out-migrants between current and previous population

POPULATION CHANGE - AUSTRALIA

				<u>Natural increase</u>				<u>Net migration</u>		
5,382,000	=	15,178,000	+	242,000	-	113,000	+	93,000	-	18,000
Population		Population		Births		Deaths		Immigration		Emigration
1983		1982		1982-1983		1982-1983		1982-1983		1982-1983

POPULATION CHANGE - AUSTRALIA

- **NATURAL INCREASE**
- **The population can sometimes change only through the process of natural increase.**
- **This refers to surplus (or deficit) of births in a population in a given time period.**
- **Natural increase = Births- Deaths or B-D**
- **= 242,000 - 113,000**
- **= 129,000**

POPULATION CHANGE - AUSTRALIA

- **NET MIGRATION**

- Net migration shows the net effect immigration and emigration on an area's population, expressed as an increase or decrease in a given period.
- Net migration=Number of immigrants - Number of emigrants or (I-O)

= 93,000 - 18,000

= 75,000

- The contribution of migration to Zambia's total population is negligible.

POPULATION CHANGE

- **MEASUREMENT OF THE POPULATION GROWTH RATE**
- **The population growth rate is the rate at which a population increases or decreases in a given year due to natural increase and net migration.**
- **This is expressed as a percentage of the base population.**
- **Thus, the population growth rate considers all components of population growth rate: births, deaths, and migration.**
- **Zambia's population growth rate between 1980 and 1990 was 3.2 percent.**

POPULATION CHANGE

- The exponential growth formula is used to measure population growth.
- $P_t = P_0 e^{rt}$
- Where,
- P_t = is current population
- P_0 = previous population
- e = exponent
- r = population growth rate
- t = time between or the number of years between P_0 and P_t

POPULATION CHANGE

- Solving for r , gives the population growth rate between two points in time:
- $r = ((\ln(P_t / P_0) / t) * 100$

POPULATION CHANGE

- In the case of Zambia, the following populations in 1980 and 1990:
- 1980 = 5,661,801
- 1990 = 7,759,161
- Therefore, the population growth rate was:
- $r = (\ln(7,759,161 / 5,661,801) / 10) * 100$
- = 3.15 %

POPULATION CHANGE

- THE RATE OF NATURAL INCREASE
-
- The rate of natural increase can be calculated as difference between birth rate (CBR) and the death rate (CDR) divided by the number of years in the intercensal period.
- Between 1980 and 1990 , Zambia's CBR was 49.9 and the CDR is 19.6
- Rate of natural increase = $\frac{\text{Birth rate} - \text{Death rate}}{\text{Number of years in interval}}$
 - = $(49.9 - 19.6)/10$
 - = 3.03
- This means that in 1990, Zambian rate of natural increase was 3.03 percent.

POPULATION CHANGE

- Since the overall annual population growth rate was 3.2, population change due to net migration is calculated as follows:
- This is simply calculated as follows:
- Net migration = Population Growth Rate – Rate Of Natural Increase.
 - $= 3.15 - 3.03$
 - $= 0.12$
- About 0.12 percent of this growth was due to migration; and 3.03 percent due to natural increase.

POPULATION CHANGE

- DOUBLING TIME OF THE POPULATION
- One of Malthus famous hypotheses was that the population would be doubling every 25 years.
- Just how can this be calculated?
- If we know the growth rate, we can estimate the doubling time of any population can be found by simply transforming of the exponential formula for population growth.
- $2P_0 = P_0 e^{rt}$
- $t = \ln(2P_0/P_0)/r$
- Or more simply, this can be estimated by dividing the rate of growth into 70.
- For example, since Zambia's rate of growth is 3.1, then the doubling time is: $70/3.1 = 23$ years.

SOCIAL AND ECONOMIC CONSEQUENCES OF POPULATION CHANGE

HIGH FERTILITY AND LOW MORTALITY

Population size	Large population size
Population composition	Increase in the size of the younger population; median age below 20
Population change	Faster and rapid population growth rate; population growth above one.

HIGH FERTILITY AND LOW MORTALITY

TECHNOLOGICAL	Young more dynamic population is more adaptable to new technological trends; it may serve vanguard for development	Abuse of technology such as cybercrime may increase, resulting in increase conflicts and tensions
POLITICAL	A more open and democratic society	If levels of education and knowledge are low, benefits of an open society not realizable.

HIGH FERTILITY AND LOW MORTALITY

	POSITIVE	NEGATIVE
SOCIAL	Demographic dividend: Younger, highly educated skilled and healthy population.	Mismatch between large younger population and employment opportunities leads to high youth unemployment: if not matched with economic growth may lead to social tensions
ECONOMIC	Large market size and economies of scale; Demographic dividend may translate into rapid economic development.	The increase in population size reduces the GDP per capita, particularly if the population growth rate exceeds the economic growth rate. If not productive, a large and dependent population may lead to more consumption and less

LOW FERTILITY AND LOW MORTALITY

Population size	Small population size
Population composition	Decrease in the size of the younger population; aging of the population; median age above 20
Population change	Slow and less rapid population growth rate; zero population growth

LOW FERTILITY AND LOW MORTALITY

	POSITIVE	NEGATIVE
SOCIAL	More worldly wisdom and institutional memory available?	This may retard development. In some cases, this may lead to increase in the tax rates for working population who may have to subsidize social security benefits for the older population. Problems of old age dependency may increase. An older population may also retard the social mobility prospects of the younger generation
ECONOMIC	The increase in size of the population is negligible. It makes it easy to achieve development goals. The GDP per capita is higher; the level of savings is higher, and there are enough investable surpluses to further boost production; there is less population density, and more space. Living standards are likely to improve.	This may reduce the productive capacity of the population because of the smaller size of the economically active population; it may result in the importation of labour; the market size reduces; the economies of scale diminish.

LOW FERTILITY AND LOW MORTALITY

TECHNOLOGICAL	The population is older and not easily adaptable to new technology, resulting in technological lag.	
POLITICAL		A less open and less democratic dispensation may result with the possibility of an emerging gerontocracy?